An AI Ethical Framework for Data-Centric Engineering

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0.1	June 14, 2026	Initial drafting, scope definition, and foundational principle mapping.	J. Frusher
0.2	June 18, 2026	Integration of the operational lifecycle gates and tier architecture.	J. Frusher
0.3	June 22, 2026	Comprehensive revision to eliminate socio-technical transience; abstracted risk tiers around analytical tractability and verification.	J. Frusher
0.4	June 24, 2026	Revised the framework's mentions of current AI capabilities to ensure the document remains relevant.	J. Frusher

Abstract

The rapid integration of artificial intelligence into safety-critical engineering systems reveals a key regulatory gap: current ethical and governance frameworks for AI are rooted in software-centric paradigms that fall short when algorithmic failure carries direct physical, clinical, or infrastructural consequences. This article proposes an ethical framework geared towards engineers, developed within the particular setting of Data-Centric Engineering (DCE). The framework is structured around five core engineering principles: Verification, Traceability, Ultimate Liability, Boundary Clarity, and Stewardship, each conceived as a professional imperative binding the practising engineer rather than the algorithm. To ensure these principles are operationalised beyond philosophy, the framework introduces a proportional, three-tiered risk classification that scales compliance obligations to the stakes of each deployment context. These tiers underpin three sequential phases of implementation: an algorithmic risk gate at project inception, automated compliance checks embedded within **CICD!** (**CICD!**) pipelines, and a structurally decoupled runtime architecture mandating deterministically safe fallback and Human-in-the-Loop (HITL) override for safety-critical systems. The framework further argues that effective governance requires a corresponding cultural shift in academic training, conditioning practitioners to treat model outputs as conditional probability distributions requiring validation rather than as verified truths. The overarching principle is that professional accountability cannot be delegated to a computational process.

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1 Introduction and Context

1.1 Background and Institutional Context

This framework represents one of the key outputs of this research project, providing a formal ethical framework for the use of Artificial Intelligence (AI) in Data-Centric Engineering (DCE). It serves as an authoritative model providing a baseline for ensuring that deployed computational pipelines are verifiable, transparent, and traceable. Taking into account the outstanding speed of current technological advancements, it is crucial to acknowledge that the proposed framework is designed to function as a living document, which can be iterated upon dynamically. It is critical to take preemptive measures due to the emergence of socio-technical risks associated with new types of engineering [1].

1.2 The Engineering Gap in AI Ethics

While established frameworks such as the Institute of Electrical and Electronics Engineers (IEEE) Global Initiative on Ethics of Autonomous and Intelligent Systems [2], International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) 42001 [3], the European Commission's Ethics Guidelines for Trustworthy AI [4], and the EU Artificial Intelligence Act [5] provide essential governance structures for data fairness and AI management systems, they are rooted in a software-centric paradigm. They do not address the cyber-physical realities codified in traditional engineering safety standards such as IEC 61508 [6] or ISO 13849 [7]. Furthermore, professional engineering codes of ethics, such as those of the American Society of Mechanical Engineers (ASME) [8] and American Society of Civil Engineers (ASCE) [9], ground liability in licensed professional practice, a principle this framework extends to AI pipeline authorship. It is the aim of the proposed framework to bridge this vital regulatory gap.

1.3 Terminology and Definitions

The following terms are used with specific technical meaning throughout this document. Precise definitions are necessary to prevent the semantic ambiguity that commonly undermines the practical application of ethical frameworks in engineering contexts.

Artificial Intelligence (AI). As used in this framework, AI denotes any computational system that employs a statistical model, trained on empirical data, to perform prediction, classification, optimisation, or decision-support tasks. This definition explicitly excludes deterministic, rule-based algorithms, which carry conventional engineering liability without the additional epistemic risks addressed here [10].

Data-Centric Engineering (DCE). An engineering paradigm in which the data pipeline, its provenance, representativeness, and integrity, is treated as the primary engineering artefact, rather than the model architecture itself. DCE recognises that the performance and safety envelope of any AI system is fundamentally bounded by the quality and scope of its training data.

Algorithmic Pipeline. The end-to-end sequence of computational processes through which raw input data is transformed into a model output or engineering recommendation. This encompasses data ingestion, pre-processing, model inference, post-processing, and any downstream actuation or advisory stages [11].

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Out-of-Distribution (OOD) Data. Input data whose statistical characteristics fall outside the distribution represented by the model's training dataset. OOD inputs invalidate the implicit assumptions of a trained model and constitute a primary and frequently underestimated failure mode in deployed AI systems [12].

Human-in-the-Loop (HITL). A system architecture or operational protocol in which a qualified engineer is required to review, validate, or explicitly approve algorithmic outputs before consequential action is taken. HITL mechanisms are the primary instrument through which nondelegable professional liability is preserved at runtime [13].

Failure Mode and Effects Analysis (FMEA). A structured, systematic methodology for enumerating the potential failure modes of a system, assessing their root causes and downstream consequences, and quantifying associated risk [14]. This framework extends the conventional FMEA to encompass probabilistic and adversarial failure modes specific to AI systems, referred to herein as a *model-focused FMEA*.

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2 The Five Core Engineering Principles

Principle I — Verification

You shall not accept any algorithmic predictions or outputs as being verified truths. All data models, sensitivities, optimisations, and simulations must be constrained within known engineering, statistical, and physical limits, and rigorously validated against empirical reality before use.

Principle II — Traceability

Provenance of training datasets and generated data must be evident to ensure the pipeline is fully reproducible and auditable. For third-party or open-source models where raw provenance is inaccessible, a formal secondary data attestation documenting known dataset limitations is required. Misrepresenting the source of data, concealing algorithmic bias, or promoting unverified AI-generated content as your own engineering design is unethical.

Principle III — Ultimate Liability

The algorithm does not possess professional liability; only the engineer does. You have ultimate, nondelegable liability for the physical and clinical failure modes of any system powered by an AI pipeline if you certify it.

Principle IV — Boundary Clarity

It is your responsibility to define explicitly where the training distribution ends and extrapolation begins using mathematically defensible metrics (e.g., conformal prediction intervals or reconstruction error thresholds). You must foster an environment in which model anomalies, distributional drift, and OOD conditions are systematically detected and disclosed.

Principle V — Stewardship

In fields where technology is advancing rapidly, you must anticipate future failure modes, adversarial vulnerabilities, and the potential for misuse or dual-use of your systems. Where there exists a credible risk of serious harm to human beings or the natural environment, the precautionary principle shall govern.

3 Operationalising the Framework

To avoid the danger of keeping this framework purely philosophical in nature, it is imperative that these principles be operationalised into concrete engineering processes. This involves incorporating verification checkpoints into the technical system.

3.1 Proportional Governance Using Tiers of Risk

To ensure institutional feasibility, compliance constraints are applied proportionally via a three-tiered risk framework rather than a blanket mandate [5], [15].

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Table 1: Data-Centric Engineering Risk Classification and Governance Matrix

Risk Tier	Verification Boundary & Operational Profile	Mandatory Governance & Validation Requirements
Tier 1: Low	Computational pathways are entirely deterministic and transparent; the practitioner possesses the analytical capacity to independently audit, reproduce, and validate the specific mechanisms generating the output.	Exempt from formal external oversight. The practitioner is required to maintain comprehensive documentation detailing foundational assumptions, data lineage, and validation methodologies within established institutional records.
Tier 2: Medium	System complexity or dimensionality prevents direct, step-by-step cognitive auditing of individual internal operations, requiring the practitioner to evaluate system performance through aggregate behavioural characteristics and bounded validation metrics.	Mandatory quantification of operational margins and uncertainty; formalised procedures for expert intervention and qualitative appraisal; documented internal technical verification prior to downstream utilisation.
Tier 3: High	System execution exhibits complete operational opacity or inherent nondeterminism, preventing the practitioner from predicting or validating the accuracy of specific outputs prior to operational deployment and systemic integration.	Implementation of independent, structurally isolated boundary constraints capable of overriding system execution; exhaustive evaluation under simulated boundary conditions; formal acknowledgement explicitly accepting the epistemic risk of deploying unverified computational outputs.

3.2 Integration into the Engineering Lifecycle

Risk boundaries must be established prior to resource allocation or software development. At this initial juncture, engineers must delineate the system's operational domain and execute a model-oriented Failure Modes and Effects Analysis (FMEA). Addressing these constraints early exploits a core principle of engineering economics: design adjustments are least disruptive and lowest in cost during the initial requirements phase [16]. During the development phase, verification shifts from a manual design task to an automated runtime checkpoint. This mechanism triggers on every code commit or dataset modification. The automated test suite must continuously validate data provenance, enforce physical boundaries, and audit for statistical dataset shift [17]. This function operates as a strict, binary gate, any verification failure must programmatically halt the integration pipeline. The final architectural determinations occur when mapping the system's control topology prior to live deployment. Designers must isolate the probabilistic model from direct physical actuation. Control authority is assigned strictly to an independent, deterministic safety controller [18]. If an anomaly or out-of-distribution input occurs, this layer autonomously brings the system to a known safe state, leaving recovery exclusively to manual human-in-the-loop (HITL) authorization.

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3.3 Academic Culture and Research Training

Technical countermeasures remain insufficient without an underlying institutional culture of systematic scepticism. Academic curricula and research training protocols must condition practitioners to treat computational outputs not as objective verities, but as complex conditional probability distributions requiring rigorous empirical validation [19].

3.4 Practical Guidelines for Effective AI Use

AI is appropriate when the output is verifiable; AI must be seen as a tool and when used correctly it can accelerate workflow and handle tasks; however, it must never be seen as a substitution for a human or a human's judgement. Efficient and Effective Use (EEU) of AI is possible and encouraged both in industry and academia, and stems from being able to determine where and when to use it and understanding the cost of using it. Costs of AI can lie within the time to evaluate its output, the ethics of misattribution of computationally generated material as human scholarship and the epistemic risk of allowing AI to make inferences that require a human's judgement. Both students and faculty are capable of EEU and should be encouraged in these safe practices, but they should remain perpetually aware of the risks of stochastic artefacts, bias and unquantified uncertainty present in AI systems [19], [20].

4 Conclusion

Professional accountability cannot be delegated to a computational process. While statistical models and automated workflows accelerate development, the responsibility for certifying that a system is safe, bounded, and fit for deployment remains an exclusively human engineering obligation.

By structuring governance into chronological, operational gates—pre-investment risk identification, automated pipeline verification, and deterministic architectural decoupling—this framework translates abstract ethical principles into verifiable design constraints. This structural approach ensures that probabilistic components function strictly within safe margins, allowing engineers to leverage the efficiencies of data-centric methods without compromising public safety or physical system integrity.

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